# Key concepts

### Stat 154/254: Statistical Machine Learning

**Note: this needs to be updated for Fall 2026**

For study purposes, this is a list of keywords that I consider central to the course. By the end of the course, you should have a good sense of what each word means and how it relates to other key concepts.

## High level concepts

Everything we do in the course is related to at least one of these connected topics.

- "Statistics"
- "Machine learning"
- "Correct specification"
- "Prediction"
- "Inference"
- "Loss"
- "Risk"
- "Generalization"
- "Overfitting"
- "Regularization"
- "Complexity"
- "Probabilistic models"
- "Computation"

## Intermediate level concepts

These topics provide detail and specificity for the high level concepts in the context of the present course. They provide a conceptual bridge between the high level topics and the actual theoretical and practical tools of the lower level topics.

- “Empiricial distribution”
- “Population distribution”
- “Empiricial risk”
- “Population risk”
- “Expected loss”
- “IID data”
- “Regression model”
- “Probability model”
- “Classifier model”
- “Overfitting”
- “Bias / variance tradeoff”
- “Conditional expectation”
- “Complexity”
- “Smoothness”
- “Basis functions”

## Lower level concepts

These are instances of specific key theoretical or practical results from the course. If you actually solve a problem or justify a method, you’ll end up using something at this level of specificity.

- “Linear models”
- “Linear regression”
- “Squared error”
- “Mean absolute error”
- “Zero-one loss”
- “Cross entropy loss”
- “Logistic regression”
- “Logistic regression loss”
- “Residual sum of squares”
- “Variable selection”
- “Law of large numbers”
- “Central limit theorem”
- “Uniform law of large numbers”
- “Indicator functions”
- “Splines”
- “Nonlinear feature maps”
- “L2 (ridge) regularization”
- “L1 (lasso) regularization”
- “Regressor colinearity”
- “Existence of unique minimizers”

- “Series expansion”
- “Taylor series”
- “Fourier series”
- “Cross validation”
- “Test set”
- “Leave–one–out cross validation”
- “K–fold cross validation”
- “True positives and false positives”
- “True negatives and false negatives”
- “Receiver operating characteristic curves”
- “Combining estimators”
- “Separating hyperplanes”
- “Sparse regression vectors”
- “Optimization duality”
- “Generative modeling for classifiers”
- “Discriminative classifiers”
- “Shattering”
- “VC dimension”
- “Integrable Lipschitz condition”
- “Maximum likelihood estimation”
- “The CART algorithm”
- “Decision trees”
- “Partitions”
- “Cost-complexity pruning”
- “Variable importance”
- “Bootstrap”
- “Bagging”
- “Generalized additive models”
- “Weak learners”
- “Boosting”
- “Gradient boosting”
- “Greedy optimization”
- “Gradient descent”
- “Stochastic gradient descent”
- “Forward stagewise optimization”

## Kernel methods (not covered yet)

- “Inner products”
- “Dot products”
- “The kernel trick”
- “The push-through identity”

- “Support vector machines”
- “Positive definite kernel”
- “Gram matrix”
- “Eigenvalues”
- “Eigenvectors”
- “Eigenfunctions”
- “Feature maps”
- “Monomial features”
- “Polynomial kernel”
- “RBF kernel”
- “Dot product kernel”
- “Reproducing kernel Hilbert space”
- “RKHS norm”
- “RKHS inner product”
- “The reproducing property”
- “Mercer’s theorem”
- “The representer theorem”
- “Inner products of functions”
- “Infinite-dimensional vector space”

## Study ideas

Here are some ways you can use this keyword list as a study aid.

- Put each topic in one or several units of the course
- Look at a given homework problem.
  - Which topics does it touch on at each level?
  - How does the homework problem connect different topics?
- Pick a low level topic and trace its “conceptual geneology” up through the intermediate and higher level concepts
- Make a “concept map” for some set of topics, drawing lines between connected ideas. Explain your concept map to someone else